Tahmidul Bin Ferdous

Chittagong, Bangladesh | tahmidul.ferdous.edu@gmail.com | +880163-0620988 | tahmidulferdous.github.io
GitHub: tahmidulferdous | LinkedIn: tahmidul-kashfi | Google Scholar | ORCID: 0009-0001-1459-4124

Education

BSc in Computer Science & Engineering *2022 – 2026*
International Islamic University Chittagong (IIUC), Chittagong, Bangladesh
Undergraduate Thesis: "Detecting Data Leakage in Lung Cancer CT Classification, across CNN, EfficientNet, and Vision Transformer models."
GPA: 3.31 / 4.00

Technical Skills

- **Programming Languages:** Python (Primary, 250+ logged experiments), C++, SQL, Bash/Shell, LaTeX.
- **Deep Learning Frameworks:** PyTorch (Advanced), JAX/Flax, torchvision, timm, TensorFlow/Keras, Hugging Face.
- **Computer Vision & ML Libraries:** OpenCV, MediaPipe, Ultralytics YOLOv11, scikit-learn, Albumentations, NumPy, Pandas, Matplotlib.
- **Systems & Infrastructure:** Distributed Data Parallel (`torchrun`, multi-GPU DDP), Google Cloud TPU v5e (8-device JAX), Mixed Precision (AMP), Weights & Biases (250 tracked runs), Git, Docker, Linux/Ubuntu.
- **Models Built from Scratch:** Autoregressive Decoder Transformer (GPT-1), Rectified Flow / Flow Matching with ODE solvers, Conditional DDPM, WGAN-GP (gradient penalty & critic loop), DCGAN, Mamba-Attention hybrid (DSMaT).

Research & Professional Experience

EliteLab *Remote*
Student Researcher *2026 – Present*
Outlier AI *Remote*
AI Data Quality & Alignment Specialist *November 2024 – February 2025*

- Audited and curated large-scale human preference and reasoning datasets utilized for Reinforcement Learning from Human Feedback (RLHF) and alignment.
- Refined multi-turn annotation guidelines, directly mitigating prompt ambiguity and improving inter-annotator reliability across reasoning benchmarks.

Personal Projects

Data Leakage in Lung Cancer CT Classification (Undergraduate Thesis) *Sep 2025 – Jun 2026*
PyTorch, timm, CCAL, EfficientNet B0–B4, ViT-B/16, Weights & Biases (188 runs) | *Code*

- Audited slice-level data splits on the IQ-OTH/NCCD lung CT dataset across four architecture families (CCAL dual-CNN attention, EfficientNet B0–B4, ViT-B/16, and a PLOS ONE four-block CNN), training every model twice under a shuffled and a file-order split with everything else held fixed.
- Showed that shuffled slice splits inflate test accuracy by $\sim$**7.5 points** at a matched split ratio over three seeds (CCAL 95.8% $\rightarrow$ 88.3%; EfficientNet 99.3% $\rightarrow$ 91.7%).

Conditional Rectified Flow for CT Synthesis on Google Cloud TPU v5e *Feb 2026 – Mar 2026*
PyTorch, JAX/Flax, Google Cloud TPU v5e, Weights & Biases (18 runs, 32.1 GPU-hours) | *Code*

- Implemented class-conditional Rectified Flow (flow matching) from original mathematical literature: linear interpolant trajectory, velocity-prediction objective, logit-normal timestep sampling, and Euler/midpoint ODE numerical solvers.
- Built multi-GPU DistributedDataParallel (DDP) PyTorch training infrastructure, then **ported the complete architecture to JAX/Flax for 8-device Google TPU v5e**, achieving **576 images/sec** (2.50 s/epoch vs. 15.5 s/epoch on 2$\times$ Tesla T4)—a $\sim$**6$\times$ throughput speedup**.

BanglaPoemGPT — 10M-Parameter Autoregressive Transformer from Scratch *March 2025*
Python, PyTorch, Kaggle P100 (4 runs, 2.0 GPU-hours) | *Code*

- Engineered a GPT-1 architecture from first mathematical principles without high-level library wrappers: multi-head causal self-attention, learned positional embeddings, custom tokenizer, and autoregressive generation loop.
- Trained on a 3M-token Bengali poetic corpus, achieving validation cross-entropy loss of **1.693** with coherent generative sampling.

Conditional DDPM for Minority-Class CT Balancing *February 2026*
PyTorch, DistributedDataParallel, 2$\times$ Tesla T4 (5 runs, 4.2 GPU-hours) | *Code*

- Constructed a class-conditional Denoising Diffusion Probabilistic Model (DDPM) featuring a conditional U-Net backbone, cosine noise schedule, and multi-GPU DDP orchestration, reaching a training loss of **0.0238** over 3,000 epochs.

Edge Vision: YOLOv11 Campus Security & Facial Biometric Systems *Oct 2024 – Nov 2024*
YOLOv11, OpenCV, FaceNet, PyTorch | *Code, Kaggle* | *Demos: ID Card Detection, Facial Attendance*

- Curated, annotated, and trained custom YOLOv11-s/m models for campus ID compliance monitoring, reaching **mAP50 = 0.82**.

- Implemented edge facial recognition utilizing SSD face detectors and FaceNet 128-d embeddings with cosine similarity matching, calibrating decision thresholds against false-accept/false-reject trade-offs.

Generative Adversarial Networks: Bengali Digits & Faces *Nov 2024*

PyTorch, DCGAN, WGAN-GP, Weights & Biases (42 runs) | *Code (Digits), Code (Faces)*

- Implemented DCGAN and WGAN-GP (gradient penalty, critic loop) from scratch to generate handwritten Bengali digits and face images, comparing training stability across adversarial objectives.

Sketch AI: Real-Time Air Canvas *Nov 2024*

MediaPipe, OpenCV, Python | *Demo*

- Built a real-time air-canvas that tracks fingertip landmarks from a webcam to draw on screen with hand gestures.

Face Mouse: Hands-Free Cursor Control *Aug 2024*

MediaPipe, OpenCV, PyAutoGUI | *Demo*

- Built a hands-free mouse for users with motor disabilities: nose landmarks drive the cursor and eye blinks trigger clicks.

Certifications

- **DeepLearning.AI Specializations (Coursera):** Neural Networks & Deep Learning, Improving Deep Neural Networks (Tuning, Regularization, Optimization), Convolutional Neural Networks, Sequence Models.

Academic References

Dr. Md. Khaliluzzaman
Associate Professor
Department of Computer Science and Engineering
International Islamic University Chittagong (IIUC)
Chittagong, Bangladesh
Email: khalil@iiuc.ac.bd

Dr. Abu Nowshed Chy
Assistant Professor
Department of Computer Science & Engineering
University of Chittagong (CU)
Chittagong, Bangladesh
Email: nowshed@cu.ac.bd